

Heart Disease Prediction Project (Logistic Regression)

1 Project Overview

This project is a machine learning-based system designed to predict the presence of heart disease in a patient based on medical attributes. It uses a supervised learning approach with Logistic Regression to classify whether a patient is likely to have heart disease or not.

2 Problem Statement

Heart disease is one of the leading causes of death worldwide. Early detection is important for prevention and treatment. The goal of this project is to build a predictive model that can classify patients based on their medical data.

3 Dataset Description

The dataset contains medical records of patients with the following features:

- age
- sex
- chest pain type (cp)
- resting blood pressure (trestbps)
- cholesterol level (chol)
- fasting blood sugar (fbs)
- resting ECG results (restecg)
- maximum heart rate achieved (thalach)
- exercise induced angina (exang)
- ST depression (oldpeak)
- target (0 = no disease, 1 = disease)

4 Data Preprocessing

The following preprocessing steps were applied:

- Removal of duplicate records
- Handling of missing values
- Separation of features (X) and target (y)
- Feature scaling using StandardScaler
- Train-test split for model evaluation

5 Model Used

Logistic Regression (Supervised Learning Algorithm)

Logistic Regression is used for binary classification problems and predicts the probability of heart disease using the sigmoid function.

6 Model Training

- The dataset was split into training and testing sets
- The model was trained on scaled training data
- Predictions were made on the test data

7 Model Evaluation

The model was evaluated using:

- Accuracy Score: 79%
- Precision, Recall, F1-Score
- Confusion Matrix

Key Results:

- Good recall for class 1 (heart disease detection)
- Balanced performance across both classes
- Effective for medical classification tasks

8 Conclusion

The Logistic Regression model successfully predicts heart disease with around 79% accuracy. The model performs well in identifying patients with heart disease, which is critical for healthcare applications.

9 Future Improvements

- Use advanced models like Random Forest and XGBoost
- Perform hyperparameter tuning
- Add more medical features
- Deploy model using Flask or Streamlit for real-time prediction